

Name of Faculty	:	Faculty of Engineering & Technology
Name of Program	:	Bachelor of Technology (B. Tech)
Course Code	:	5BIT02
Course Title	:	Web Programming
Type of Course	:	Professional Core (PC)
Year of Introduction	:	2024-25

Prerequisite	:	Knowledge of basic HTML is required.
Course Objective	:	This course aims to provide students with a thorough understanding of the fundamental Web Development used in computer science. The focus is on teaching how to efficiently organize, manage, and manipulate data using various data structures. Students will learn to evaluate and select appropriate Web Development based on the requirements of the problem at hand, thus enhancing their problem-solving skills and understanding of web Development efficiency.
Course Outcomes	:	At the end of this course, students will be able to:
	CO1	Understand the concepts of WWW, HTTP protocol and client-server architecture
	CO2	Learn and apply various HTML tags to build the user friendly web pages
	CO3	Explore the new features of CSS to define and apply CSS rules in the web pages for rich User Interface.
	CO4	Create interactive web pages to improve the user experience using client side scripting with JavaScript.
	CO5	Design and develop fully functional dynamic web applications using the concepts of PHP, MySQL

Teaching and Examination Scheme

Teaching Scheme (Contact Hours)			Credits	Examination Marks				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	SEE	CIA	SEE	CIA	
3	0	2	4	50	50	25	25	150

Legends: L-Lecture; T-Tutorial/Teacher Guided Theory Practice; P-Practical, C-Credit, SEE-Semester End Examination, CIA-Continuous Internal Assessment (It consists of Assignments/Seminars/Presentations/MCQ Tests, etc.))

Course Content

Unit No.	Topics	Teaching Hours	Weightage	Mapping with CO
1	Introduction: Basics of WWW, HTTP protocol methods and headers, HTTP Request and Response, Architecture of web browser, Web server installation and configuration, Web security, CORS, Understanding SEO	2	10%	CO1
2	HTML & CSS: HTML page structure, formatting tags in HTML, tables, links, images, meta tags, frames, html form tags, media, APIs, HTML5 tags in relation to validations and SEO. CSS: Need for CSS, Basic syntax and structure, Backgrounds, Colors and properties, Manipulating texts, Fonts, borders and boxes, Margins, Padding Lists, CSS2, CSS3, Animations, Tool-Tips, Style images, Variables, Flex Box, Media Queries, Wildcard Selectors (*, ^ and \$) in CSS, Working with Gradients, Pseudo Class, Pseudo elements, basic of frameworks like Bootstrap, Responsive web design and Media Query, CSS variables	08	30%	CO2
3	Java Script: JavaScript Syntax, Types of JavaScript, variables, arrays, functions, conditions, loops, Pop up boxes, JavaScript objects and DOM, JavaScript inbuilt functions, JavaScript validations, Regular expressions, Event handling with JavaScript, Callbacks in JavaScript, Function as arguments in JavaScript, Object concepts in JavaScript, JSON	10	30%	CO2 CO3
4	PHP Basics Introduction to Server side programming, PHP variables, decision and looping with examples, PHP and HTML, Arrays, Functions, Browser control and detection, String, Form processing, File uploads, Dates and time zone, working with Regular Expressions, Exception Handling, Working with JSON data, Object Oriented Programming with PHP Session and State Management using PHP Need of session management, Various techniques for state and session management like: Hidden Fields, Query String, Cookie and Session	10	15%	CO4
5	Database Connectivity using PHP: Basic commands for database	5	15%	CO5

	connection and query execution with CRUD examples, Object oriented database access using PHP			
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Suggested Distribution of Theory Marks Using Bloom's Taxonomy						
Level	Remembrance	Understanding	Application	Analyse	Evaluate	Create
Weightage	10	20	25	10	10	15

NOTE: This specification table shall be treated as a general guideline for the students and the teachers. The actual distribution of marks in the question paper may vary slightly from above table.

Suggested List of Experiments/Tutorials

Sr. No.	Name of Experiment/Tutorial	Teaching Hours
1	Install and configure a web server (XAMPP/WAMP/LAMP) and test basic HTML and PHP pages.	2
2	Create a basic web page with proper structure using HTML5 tags.	2
3	Design a responsive web page using CSS Flexbox and Grid Layout.	2
4	Use Bootstrap framework to create a responsive multi-page website.	2
5	Write a JavaScript program to demonstrate variables, arrays, and functions.	2
6	Implement form validation using JavaScript and regular expressions.	2
7	Implement JavaScript event handling for mouse and keyboard events.	2
8	Demonstrate PHP arrays and functions for data manipulation.	2
9	1. Establish a MySQL database connection using PHP. 2. Perform CRUD (Create, Read, Update, Delete) operations on a database using PHP and MySQL.	2
10	Implement user authentication (login/logout) with hashed passwords in PHP and MySQL.	2

Suggested Learning Websites

Sr. No.	Name of Website
1	Web Security and SEO: https://www.tutorialspoint.com/seo/index.htm https://github.com/vasanthk/web-security-basics
2	HTML: https://developer.mozilla.org/en-US/docs/Web/HTML https://www.w3schools.com/html/ https://www.tutorialspoint.com/html/index.htm
3	CSS: https://developer.mozilla.org/en-US/docs/Web/CSS https://www.manning.com/books/css-in-depth

	https://www.w3schools.com/css/ https://www.tutorialspoint.com/css/index.htm
4	Java Script: https://javascript.info/ https://github.com/getify/You-Dont-Know-JS https://www.w3schools.com/js/ https://www.tutorialspoint.com/javascript/index.htm
5	PHP: https://www.w3schools.com/php/ https://www.tutorialspoint.com/php/index.htm

Reference Books

Sr. No.	Name of Reference Books
1.	HTML 5 Black Book 2Ed, by Kogent Learning Solutions Inc.
2.	Learning PHP, MySQL, JavaScript, CSS & HTML5, 3rd Edition.
3.	A Step-by-Step Guide to Creating Dynamic Websites By Robin Nixon Publisher: O'Reilly Media.
4.	JavaScript for impatient programmers by Dr. Axel Rauschmayer.
5.	PHP: The Complete Reference By Steven Holzner, McGrawhill.